

# Exogenous-Reservoir Exponential-Quadratic Volatility Model

Mathematical specification, implementation guide, and M2–M4 ablation programme

**Purpose.** This document is the typeset companion to the Python package and README. It explains the implemented equations, distinguishes mandatory from optional components, describes the effects and risks of M2, M3, and M4, and gives a controlled experimental design for deciding whether the optional layers should be retained.

|                          |                                                                    |
|--------------------------|--------------------------------------------------------------------|
| <b>Status:</b>           | executable research specification, not a market calibration result |
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## Abstract

The implemented model is a continuous-path stochastic-volatility architecture descended from ESN–A2. Exogenous multirate Ornstein–Uhlenbeck factors form a reusable linear reservoir. A normalised and smoothly tempered exponential-quadratic readout maps the reservoir to positive variance. Under the physical measure, the target inventory comprises finite-frequency heavy tails, volatility clustering, finite-scale roughness, leverage, a strong Zumbach effect, near-zero return autocorrelation, and, at lower priority, the Taylor effect. Under the pricing measure, the eventual objective is a joint fit of SPX options, VIX futures, and VIX options.

M2 is the recommended Gaussian-reservoir baseline. M3 adds a fixed contractive nonlinear echo-state feature. M4 adds both that nonlinear feature and an independent quadratic dispersion bank. The recommendation is to implement all variants as challengers but not to start the primary calibration with M3 or M4. A two-by-two factorial ablation—M2, M3, M2 plus the orthogonal bank, and M4—is more informative than the nested comparison M2 versus M3 versus M4 alone, because it separates the nonlinear and dispersion effects.

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## 1 Executive decision

### Recommendation

Use **M2 as the primary baseline**. Build M3 and M4 now as experimental challengers, but activate them only in paired ablations using the same primitive Gaussian paths, the same core parameter candidates, and independent validation windows. Test the echo-state feature and the orthogonal dispersion bank separately before testing them jointly.

This recommendation is stronger than a simple nested ladder. The scientifically clean design has two binary switches:

| Label | Echo feature | Orthogonal bank | Role                                                  |
|-------|--------------|-----------------|-------------------------------------------------------|
| M2    | off          | off             | Mandatory baseline and default production candidate   |
| M3    | on           | off             | Tests nonlinear cross-feature memory                  |
| M2+O  | off          | on              | Tests independent dispersion without nonlinear memory |
| M4    | on           | on              | Tests both additions and their interaction            |

The intermediate model M2+O is not a new production name; it is an essential ablation. Comparing only M2, M3, and M4 would confound the effect of the orthogonal block with the continued presence of the echo feature.

The optional components should be retained only if they improve high-priority diagnostics out of sample by more than Monte Carlo and sampling uncertainty, do not materially damage leverage or Zumbach, do not force frequent cap activation, and yield stable parameters away from zero and from search-box boundaries.

## 2 Research objective and hierarchy

The experiment asks whether the architecture has a *non-negligible and stable region* of parameter space that satisfies the requested facts jointly. Finding one exceptional point is insufficient. Conversely, a pass rate is meaningless unless the parameter box and the sampling measure are reported: widening an arbitrary box lowers the pass rate, while hand-selecting a narrow box raises it.

The implemented hierarchy is:

1. calibrate architecture-level latent feature shapes;
2. draw primitive Gaussian randomness once;
3. optionally sample state-affecting correlations and deterministic risk premia in an outer design;
4. sample cheap readout parameters in an inner space-filling design;
5. retain every candidate and report the joint accepted region, not only its optimum;
6. repeat over independent cache seeds and validation windows.

Only latent rough-kernel and clustering curves are close to hyperparameter-only targets. Measured return tails, leverage, Zumbach, Taylor, and post-readout roughness depend on interactions between architecture, structural parameters, tempering, and the readout.

### 3 Exogenous state system

#### 3.1 OU banks and driver correlations

Let  $W^{S,\mathbb{M}}$  drive the equity return under  $\mathbb{M} \in \{\mathbb{P}, \mathbb{Q}\}$ . The feedback and spike drivers are

$$dW_t^{F,\mathbb{M}} = \rho_F dW_t^{S,\mathbb{M}} + \sqrt{1 - \rho_F^2} dB_t^{F,\mathbb{M}}, \quad (1)$$

$$dW_t^{J,\mathbb{M}} = \rho_J dW_t^{S,\mathbb{M}} + \sqrt{1 - \rho_J^2} dB_t^{J,\mathbb{M}}. \quad (2)$$

The slow clustering bank  $C$  and optional orthogonal bank  $O$  use independent primitives by default. For bank  $b \in \{F, J, C, O\}$ , the standardised physical-measure coordinates are

$$\boxed{dX_{j,t}^{b,\mathbb{P}} = -\lambda_j^b X_{j,t}^{b,\mathbb{P}} dt + \sqrt{2\lambda_j^b} dW_{j,t}^{b,\mathbb{P}}.} \quad (3)$$

Thus every coordinate has stationary variance one. For  $N_b \geq 2$ , the default rate grid is geometric:

$$\lambda_j^b = \lambda_{\min}^b \left( \frac{\lambda_{\max}^b}{\lambda_{\min}^b} \right)^{(j-1)/(N_b-1)}. \quad (4)$$

#### 3.2 Fixed normalised projections

The feature engine forms

$$Y_t = a^\top X_t^F, \quad F_t = f^\top X_t^J, \quad C_t = c^\top X_t^C, \quad (5)$$

where the rate-power profiles are fixed at the architecture stage and normalised so that  $\text{Var}_{\mathbb{P}}(Y_t) = \text{Var}_{\mathbb{P}}(F_t) = \text{Var}_{\mathbb{P}}(C_t) = 1$  at the stationary reference point. For raw weights  $u_j = (\lambda_j)^p$  and stationary block covariance  $\Sigma_b$ ,

$$w = \frac{u}{\sqrt{u^\top \Sigma_b u}}. \quad (6)$$

Changing these projection profiles does not require new random numbers, but it changes the feature engine and belongs to architecture selection rather than the inner readout search.

#### 3.3 Economic roles of the four banks

| Bank         | Construction                                    | Intended effect                                                           | Main risk                                                       |
|--------------|-------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------|
| Feedback $F$ | Shared multirate driver, correlated with equity | Finite-scale roughness, leverage, Zumbach, and tail convexity through $Y$ | Excess curvature or correlation can dominate all other channels |

| Bank           | Construction                                              | Intended effect                                                           | Main risk                                                        |
|----------------|-----------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------|
| Spike $J$      | Fast and medium rates, partly equity-correlated           | Sharp volatility and VIX-like peaks with fast initial relaxation          | A dominant symmetric spike term can dilute leverage and Zumbach  |
| Clustering $C$ | Slow independent OU mixture, linear score contribution    | Persistent volatility autocorrelation without endogenous feedback         | Too-slow modes create near-non-identification on finite samples  |
| Orthogonal $O$ | Independent modes entering through average squared energy | Extra dispersion, finite-frequency tails, and possible Taylor improvement | Dilution of return-volatility dependence; superfluous dimensions |

## 4 Optional nonlinear echo-state layer

Let  $Z_n^{\text{vec}} \in \mathbb{R}^{N_Z}$  satisfy

$$Z_{n+1}^{\text{vec}} = (1 - \alpha)Z_n^{\text{vec}} + \alpha \tanh(AZ_n^{\text{vec}} + B\psi(X_n) + b), \quad (7)$$

where  $A, B, b$  are drawn once and held fixed. The implemented feature map contains linear states and centred squares. A sufficient echo-state contraction condition is

$$(1 - \alpha) + \alpha\|A\|_2 < 1. \quad (8)$$

The code projects this vector onto one fixed normalised direction,  $Z_n = d^\top Z_n^{\text{vec}}$ , and estimates only its scalar loading. This is a deliberately parsimonious M3: it avoids fitting a free coefficient for every reservoir unit.

### What M3 can and cannot add

M3 can encode bounded nonlinear interactions and cross-bank memory not represented by the fixed linear and quadratic directions. It is not automatically a better model. It increases architecture dependence, weakens Gaussian conditional-moment tractability, and may fit sampling noise. The echo matrices must be fixed before the readout experiment; repeatedly trying seeds and retaining the best one is itself hyperparameter optimisation.

## 5 Exponential-quadratic readout

### 5.1 Score

The implemented score is

$$\boxed{\eta_t = c_C C_t + q_Z (Y_t - \beta)^2 + q_J F_t^2 + q_O E_t^O + \delta Z_t}, \quad E_t^O = \frac{1}{N_O} \|X_t^O\|^2. \quad (9)$$

The identified roles are:

- $c_C$  controls slow exogenous clustering;
- $q_Z > 0$  controls feedback convexity and finite-frequency tail amplification;

- $\beta > 0$ , together with  $\rho_F > 0$ , orients the leverage effect;
- $q_J \geq 0$  controls fast spike convexity;
- $q_O \geq 0$  is the optional independent dispersion loading;
- $\delta$  is the optional fixed echo-feature loading.

There is no separate linear term in  $Y$ : a term  $\ell Y + q_Z(Y - \beta)^2$  would make  $\ell$  and  $\beta$  redundant up to a deterministic shift.

## 5.2 Smooth upper tempering

For cap level  $L$  and sharpness  $\kappa > 0$ , define

$$h_{L,\kappa}(x) = L - \frac{1}{\kappa} \log\left(1 + e^{\kappa(L-x)}\right), \quad G_{L,\kappa}(x) = e^{2h_{L,\kappa}(x)}. \quad (10)$$

Well below  $L$ ,  $h_{L,\kappa}(x) \approx x$ ; above  $L$ , it approaches  $L$  smoothly. Tempering controls moments and supports a finite-horizon martingale argument, but a cap that binds frequently flattens extreme spikes and can distort tails, roughness estimates, and the VIX smile. The code therefore records P and Q cap-exceedance fractions.

## 5.3 Forward-variance normalisation

Given a deterministic forward-variance input  $\xi_0(t) > \underline{v}$ ,

$$V_t = \underline{v} + (\xi_0(t) - \underline{v}) \frac{G_{L,\kappa}(\eta_t)}{M_Q(t)}, \quad M_Q(t) = \mathbb{E}_0^Q[G_{L,\kappa}(\eta_t)]. \quad (11)$$

Consequently  $V_t > \underline{v}$  and  $\mathbb{E}_0^Q[V_t] = \xi_0(t)$ . In the current research engine,  $M_Q$  is estimated cross-sectionally from a matched Q ensemble. This is suitable for physical-measure capacity experiments and path-reuse verification; conditional production VIX pricing still requires quadrature, regression, quantisation, or a controlled surrogate.

# 6 Measures, equity dynamics, and exact simulation

## 6.1 P and Q dynamics

Under  $\mathbb{P}$ , the stock follows

$$\frac{dS_t}{S_t} = (r_t - d_t + \theta\sqrt{V_t}) dt + \sqrt{V_t} dW_t^{S,\mathbb{P}}. \quad (12)$$

Use the deterministic Girsanov convention

$$dB_t^{\mathbb{P}} = dB_t^{\mathbb{Q}} - \gamma dt, \quad \gamma_S = \theta. \quad (13)$$

Then

$$\frac{dS_t}{S_t} = (r_t - d_t) dt + \sqrt{V_t} dW_t^{S,\mathbb{Q}}, \quad (14)$$

$$dX_{i,t}^{\mathbb{Q}} = \left(-\lambda_i X_{i,t}^{\mathbb{Q}} - \sqrt{2\lambda_i} \gamma_i\right) dt + \sqrt{2\lambda_i} dW_{i,t}^{\mathbb{Q}}. \quad (15)$$

For a common initial state and constant effective risk price  $\gamma_i$ ,

$$X_{i,t}^{\mathbb{Q}} = X_{i,t}^{\text{centred}} - \sqrt{2\lambda_i\gamma_i} \frac{1 - e^{-\lambda_i t}}{\lambda_i}. \quad (16)$$

Thus risk-premium changes require no new Gaussian draw. They do require recomputation of the Q mean, the Q echo feature when enabled, the normaliser, and all downstream pricing objects.

## 6.2 Exact adapted transition

Let  $\zeta_i$  be the exact OU innovation over a step  $\Delta$ , let  $R_{ij}$  be the instantaneous correlation of factor drivers, and let  $\rho_i$  be the asset loading of factor  $i$ . Then

$$\text{Cov}(\zeta_i, \zeta_j) = R_{ij} \frac{2\sqrt{\lambda_i\lambda_j}}{\lambda_i + \lambda_j} \left(1 - e^{-(\lambda_i + \lambda_j)\Delta}\right), \quad (17)$$

$$\text{Cov}(\zeta_i, \Delta W^S) = \rho_i \sqrt{2\lambda_i} \frac{1 - e^{-\lambda_i\Delta}}{\lambda_i}, \quad (18)$$

$$\text{Var}(\Delta W^S) = \Delta. \quad (19)$$

The implementation conditions the factor innovations on the first primitive equity shock. Consequently the equity Brownian path remains bitwise identical when a correlation is changed, while the affected OU states are transformed again. Variance at a fine step uses the state at the start of that step; this avoids the anticipative within-step bias present in simulations that update the state with  $\Delta W^S$  before evaluating the same-step volatility.

## 7 M2, M3, M2+O, and M4

### 7.1 M2: recommended baseline

M2 imposes  $\delta = 0$ ,  $q_O = 0$ , and uses

$$\eta_t^{\text{M2}} = c_C C_t + q_Z (Y_t - \beta)^2 + q_J F_t^2. \quad (20)$$

It retains the load-bearing mechanisms: clustering, finite-scale roughness, leverage, Zumbach, finite-frequency tails, and fast spike relaxation. Its state remains Gaussian and the only nonlinearities are identified quadratic directions and the tempered exponential map. It is the most interpretable specification and the correct benchmark for deciding whether more flexibility is necessary.

### 7.2 M3: M2 plus one nonlinear echo feature

M3 enables equation (7) and adds

$$\eta_t^{\text{M3}} = \eta_t^{\text{M2}} + \delta Z_t. \quad (21)$$

**Potential gain.** Bounded nonlinear cross-bank interactions, additional memory profiles, improvements in clustering shape, Taylor diagnostics, or option-capacity residuals that M2 cannot remove.

**Potential cost.** Seed sensitivity, greater hyperparameter burden, weaker identifiability, more expensive Q conditional expectations, and overfitting. If the fitted  $\delta$  concentrates near zero or its improvement disappears across seeds, M3 has not earned its complexity.

### 7.3 M2+O: orthogonal dispersion ablation

With echo disabled and a non-empty independent bank,

$$\eta_t^{\text{M2+O}} = \eta_t^{\text{M2}} + q_O E_t^O. \quad (22)$$

**Potential gain.** Higher log-volatility dispersion, heavier finite-frequency return tails, a larger Taylor gap, and additional VIX convexity without increasing the largest loading on the correlated feedback direction.

**Potential cost.** Because  $E^O$  is independent of returns, it can dilute correlation-based leverage and Zumbach statistics. A large  $N_O$  with average energy  $N_O^{-1} \|X^O\|^2$  also concentrates by the law of large numbers unless  $q_O$  or cross-mode dependence is adjusted. More dimensions are not automatically more dispersion.

### 7.4 M4: both optional mechanisms

M4 uses

$$\eta_t^{\text{M4}} = \eta_t^{\text{M2}} + q_O E_t^O + \delta Z_t. \quad (23)$$

M4 has the largest capacity but the weakest immediate identifiability. It should be tested only after estimating the separate effects of  $\delta Z$  and  $q_O E^O$ . A joint gain may arise from a genuine interaction; it may also arise because one optional channel compensates for a poorly specified other channel.

### 7.5 Context: M0 and M1

M0 is a direct positive quadratic-variance benchmark. M1 is the untempered Gaussian exponential-quadratic shadow model  $h(x) = x$ . They are useful falsification and analytic benchmarks, not recommended production choices. In particular, M1 requires explicit quadratic-moment and martingale-domain checks; a very high finite cap is not mathematically identical to removing the cap.

## 8 Mechanisms for the stylised facts

### 8.1 Near-zero return autocorrelation

The martingale return component over a short interval is of order  $\sqrt{\Delta}$ , whereas the predictable P drift is of order  $\Delta$ . Near-zero linear return autocorrelation is therefore structurally natural, although not an identity under  $\mathbb{P}$ . It must be checked after changing the Sharpe ratio or introducing additional state-dependent mechanisms.

### 8.2 Volatility clustering

For independent cluster coordinates,

$$\text{Cov}(C_t, C_{t+\tau}) = \sum_{j=1}^{N_C} c_j^2 e^{-\lambda_j^C \tau}. \quad (24)$$

A geometric mixture approximates slow decay over a selected finite band. M3 can modify the post-readout ACF through nonlinear memory; M2+O also adds quadratic decay modes. Neither addition should be used to repair a plainly inadequate slow rate grid.

### 8.3 Finite-scale roughness

The target is

$$\mathbb{E}\left[(\log V_{t+\tau} - \log V_t)^2\right] \asymp C\tau^{2H}, \quad H < \frac{1}{2}, \quad (25)$$

over the selected intraday-to-daily band. A finite multirate OU lift can approximate this scaling on that band. It has Brownian  $H = 1/2$  behaviour at sufficiently small asymptotic lags. M3 or M4 may change the measured finite-band slope, especially near stationary points of the readout or near cap saturation; they do not turn a finite lift into a genuinely fractional process.

### 8.4 Leverage

The feedback derivative is

$$\frac{\partial \eta}{\partial Y} = 2q_Z(Y - \beta). \quad (26)$$

Near  $Y = 0$ , the leading covariation sign is

$$\text{sign}(\text{d}\langle \log S, \log V \rangle_t) \approx -\text{sign}(q_Z \beta \rho_F). \quad (27)$$

Thus the current convention  $q_Z, \beta, \rho_F > 0$  gives negative leverage. Symmetric independent variance channels generally dilute its correlation magnitude. M3 has no guaranteed leverage sign and must be checked by ablation.

### 8.5 Strong Zumbach effect

For horizon  $\tau$ , a representative statistic is

$$Z(\tau) = \text{Corr}\left(R_{t-\tau,t}^2, \text{RV}_{t,t+\tau}\right) - \text{Corr}\left(\text{RV}_{t-\tau,t}, R_{t,t+\tau}^2\right). \quad (28)$$

The causal return-correlated OU filter and shifted quadratic readout break time-reversal symmetry without variance-dependent state noise. This mechanism already exists in M2. An independent orthogonal bank cannot create it directly and may dilute it. The empirical target must use intraday realised variance; daily squared returns are an inadequate proxy.

### 8.6 Finite-frequency heavy tails

Conditional Gaussian returns become heavy finite-frequency variance mixtures. In the untempered scalar illustration,

$$\sigma = e^{qY^2}, \quad Y \sim \mathcal{N}(0, 1) \quad \implies \quad \mathbb{P}(\sigma > x) \asymp x^{-1/(2q)}(\log x)^{-1/2}. \quad (29)$$

Tempering eventually truncates this asymptotic behaviour, so Hill estimates are diagnostic finite-range slopes rather than evidence of a true power law. M2 already has two convex quadratic channels. The orthogonal block is justified only if M2 cannot match tail quantiles and aggregation profiles without excessive feedback curvature or cap interaction.

### 8.7 Taylor effect

The Taylor effect requires

$$\text{Corr}(|r_t|, |r_{t+k}|) > \text{Corr}(r_t^2, r_{t+k}^2) \quad (30)$$

over a relevant lag set. It requires sufficient persistent log-volatility dispersion and is not implied by leverage, Zumbach, or roughness. M3 may alter the temporal dependence; M2+O may increase dispersion. Because Taylor is lower priority, neither optional component should be accepted solely to improve this statistic while damaging leverage, Zumbach, or tail stability.

## 8.8 Fast spike relaxation

For a fast state shocked to  $F_t = f_0$ , ignoring subsequent noise,

$$F_{t+s} \approx f_0 e^{-\lambda_J s}, \quad \frac{V_{t+s}}{\bar{V}_{t+s}} \approx \exp\left(2q_J f_0^2 e^{-2\lambda_J s}\right). \quad (31)$$

The log-relative-variance decay is steepest immediately after the peak and slows thereafter. This desired asymmetry is already present in M2 and should not be attributed to the echo layer.

## 9 Path reuse: exact meaning

For fixed state dimensions and rate grids, the implementation draws standard normals once. The following table states the correct reuse boundary.

| Change                                | New draw?                | Required recomputation                                                                                             |
|---------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------|
| Readout coefficient                   | No                       | Score, tempering, $M_Q$ , variance, returns, diagnostics, and payoffs                                              |
| Deterministic risk premium            | No                       | Q mean shift; Q echo if enabled; then all pricing quantities                                                       |
| Feedback or spike correlation         | No                       | Affected OU transformation and all downstream features; equity primitive stays fixed                               |
| Forward-variance curve                | No                       | Variance level, stock path, VIX functionals, and payoffs                                                           |
| Projection profile                    | No                       | Feature projection and downstream quantities                                                                       |
| OU rates, dimensions, or ESN matrices | No new seed is necessary | Full feature-engine reconstruction from compatible primitives; dimension changes require a compatible random cache |

### Important distinction

Path reuse does not mean that final  $S$ ,  $V$ ,  $VIX$ , or payoff paths remain unchanged when a volatility parameter changes. It means the primitive randomness and all parameter-independent features are reused. Reusing final stock paths after changing variance would be incorrect.

## 10 Recommended factorial experiment

### 10.1 Stage A: architecture selection

Fix candidate dimensions. Calibrate the feedback rate endpoints and projection power to a target finite-band rough variogram, and calibrate the cluster rate endpoints and profile to a target latent

ACF. Use rolling or nested validation; do not select hyperparameters and judge capacity on the same window.

The latent objective should not contain leverage, Zumbach, tails, or Taylor as if they were hyperparameter-only facts. Those belong to the full-model experiment.

## 10.2 Stage B: M2 capacity

Run a Latin-hypercube or scrambled Sobol design over the M2 readout parameters and an outer design over correlations and deterministic risk premia. Use common random numbers, multiple path-cache seeds, and at least the relevant intraday and daily aggregations.

Record the entire candidate table, including failures. Primary outputs are:

- joint acceptance rate with uncertainty across cache seeds;
- accepted-region volume under the declared sampling measure;
- connectedness and boundary concentration of the accepted set;
- Pareto trade-offs, especially tails/Taylor versus leverage/Zumbach;
- cap-binding rate, normaliser stability, and forward-mean error;
- parameter sensitivity and identifiability diagnostics.

## 10.3 Stage C: paired optional-component ablations

For each accepted and near-accepted M2 core vector  $\vartheta$ , evaluate a matched grid or space-filling design containing the exact zero loadings:

$$(\delta, q_O) \in \{(0, 0), (\delta, 0), (0, q_O), (\delta, q_O)\}. \quad (32)$$

For any scalar diagnostic or loss  $T$ , define paired effects

$$D_E(\vartheta) = T(\vartheta, \delta, 0) - T(\vartheta, 0, 0), \quad (33)$$

$$D_O(\vartheta) = T(\vartheta, 0, q_O) - T(\vartheta, 0, 0), \quad (34)$$

$$I_{EO}(\vartheta) = T(\vartheta, \delta, q_O) - T(\vartheta, \delta, 0) - T(\vartheta, 0, q_O) + T(\vartheta, 0, 0). \quad (35)$$

Common random numbers make these paired contrasts much less noisy than comparing independent simulations. The exact zero controls are important: a generic Latin-hypercube design almost surely never samples zero, so it does not by itself provide a nested-model comparison.

## 10.4 Stage D: selection rule

Keep an optional component only if all of the following hold:

1. it improves predeclared high-priority metrics or joint acceptance out of sample;
2. the improvement exceeds paired Monte Carlo uncertainty across seeds;
3. it does not materially worsen leverage, Zumbach, return ACF, or cap behaviour;
4. the loading is stable and not concentrated at zero or a search boundary;
5. the gain survives frequency aggregation and reasonable parameter-box perturbations;

6. for the eventual Q exercise, it improves option-capacity tests enough to justify the extra conditional-expectation cost.

Raw in-sample loss reduction is not a sufficient selection rule because M3 and M4 are more flexible by construction.

## 11 Mapping to the Python package

### 11.1 Default switches

The default `ArchitectureConfig` has a disabled echo layer and an empty orthogonal bank. The default `ParameterSpace.plausible_m2()` fixes both optional loadings at zero. Merely calling `plausible_m3()` or `plausible_m4()` is not enough: the corresponding feature engine must also be enabled.

```
from dataclasses import replace

from esn_eq import ArchitectureConfig
from esn_eq.config import BankConfig, EchoStateConfig, geometric_rates

base = ArchitectureConfig() # M2

m3 = replace(
    base,
    echo=EchoStateConfig(enabled=True, units=12, seed=20260901),
)

m2_plus_o = replace(
    base,
    orthogonal=BankConfig(
        rates=geometric_rates(0.5, 100.0, 12),
        asset_correlation=0.0,
        shared_driver=False,
    ),
)

m4 = replace(
    m2_plus_o,
    echo=EchoStateConfig(enabled=True, units=12, seed=20260901),
)
```

The official helpers provide M2, M3, and nested M4 spaces. For the orthogonal-only ablation, construct an M2 space with a positive  $q_O$  bound and fixed  $\delta = 0$ . This explicit construction is preferable to mislabelling the nested M4 space as an orthogonal-only test.

### 11.2 Reproducible baseline command

```
python -m pip install -e .
python -m unittest discover -s tests -p 'test_*.py' -v

esn-eq-explore \
  --samples 64 \
  --structural-samples 8 \
  --years 8 \
  --burn-years 2 \
  --paths 8 \
  --sampler latin_hypercube \
  --seed 2026 \
  --output exploration_results
```

This baseline evaluates  $8 \times 64 = 512$  joint candidates and writes:

- **candidates.csv**: parameters, diagnostic means, Monte Carlo standard errors, acceptance, distance, and numerical audit fields;
- **rank\_sensitivities.csv**: marginal Spearman screens;
- **summary.json**: candidate count and joint acceptance rate.

Spearman screens are exploratory sensitivity summaries, not causal effects. The paired factorial contrasts above are the more informative tool for the M3/M4 decision.

### 11.3 Frequency conventions

With `observations_per_year=252`, outputs are daily. For intraday output, set the desired number of observations per year and keep all OU rates in inverse years. `DiagnosticConfig` makes lag conventions explicit. The function `aggregate_result` aggregates matched fine-frequency returns and variance to coarser frequencies without re-simulating the states.

## 12 Diagnostics and interpretation

The package reports maximum absolute return ACF, mean log-variance ACF, finite-band roughness, Pearson and rank leverage, Pearson and rank Zumbach asymmetry, excess kurtosis, an absolute-return tail ratio, a multi-threshold Hill summary, Taylor gap and positive-lag fraction, and mean and 99.5 percent volatility. Estimates are computed pathwise before reporting Monte Carlo standard errors.

#### Interpretation safeguards

The Hill statistic is only a finite-range tail diagnostic because tempering ultimately truncates the mixture. A finite OU lift is rough only on its designed scale band. A positive simulated Zumbach statistic demonstrates mechanism capacity, not the empirical magnitude. Default acceptance bands and parameter ranges are placeholders, not SPX estimates.

For model comparison, report both raw diagnostics and standardised distances to empirical confidence bands. A candidate should not be accepted because a weighted scalar score allows one severe failure to compensate for several successes; joint hard constraints and a continuous secondary distance should be shown together.

## 13 What is optional and when to activate it

| Component      | Default            | Activate when                                                                 | Reject or defer when                                              |
|----------------|--------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Tempering      | Mandatory in M2–M4 | Production and robust physical experiments                                    | Only remove in the analytic M1 shadow with explicit domain checks |
| Echo layer $Z$ | Disabled           | Stable paired gains reveal missing nonlinear memory or cross-bank interaction | Gain is seed-specific, in-sample only, or $\delta \approx 0$      |

| Component                | Default                 | Activate when                                                                     | Reject or defer when                                                               |
|--------------------------|-------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Orthogonal bank<br>$O$   | Empty, $q_O = 0$        | M2 tails or Taylor remain deficient after cap and correlated channels are checked | It dilutes leverage/Zumbach or energy concentration makes it ineffective           |
| Joint M4                 | Disabled                | Separate echo and orthogonal effects are both positive, with a stable interaction | One component merely compensates for the other or parameters are weakly identified |
| Endogenous activity bank | Not in current baseline | Exogenous slow modes cannot reproduce empirical clustering out of sample          | It breaks strong frozen-feature reuse and adds readout-dependent recursion         |

## 14 Limitations and next decisions

### Current scope

The package is a physical-measure capacity and path-reuse engine, not yet a production joint SPX/VIX pricer. No option surface has been calibrated. Joint Q capacity remains an open empirical and numerical question.

The next decisions should be made in this order:

1. replace illustrative physical bands with estimates and uncertainty from matched intraday and daily data;
2. run the replicated M2 capacity experiment;
3. add the two separate optional-component ablations M3 and M2+O;
4. test M4 only if at least one separate effect is material;
5. construct synthetic SPX/VIX surfaces from strong benchmark models and test Q capacity;
6. only then decide whether a conditional-expectation surrogate is needed for production VIX pricing.

The defensible conclusion at this stage is architectural feasibility, not simultaneous market fit. M2 is already rich enough that failure should first trigger a diagnosis of rate grids, measurement conventions, cap placement, or target incompatibility. Optional complexity should be a response to a demonstrated residual deficiency, not the starting point.

## 15 Compact experiment checklist

1. Freeze data definitions, frequency, lag windows, and empirical confidence bands.
2. Freeze architecture candidates and echo seeds before readout exploration.
3. Include exact zero controls for  $\delta$  and  $q_O$ .
4. Reuse primitive paths and core candidates across all four factorial cells.

5. Repeat across independent cache seeds and rolling validation windows.
6. Report pass rates, accepted-region geometry, paired effects, and cap diagnostics.
7. Treat Taylor as secondary; prioritise tails, roughness, clustering, leverage, and Zumbach.
8. Reject optional layers that do not produce stable out-of-sample gains.
9. Keep Q conclusions separate until synthetic or real option surfaces are tested.

## Conceptual references

## References

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